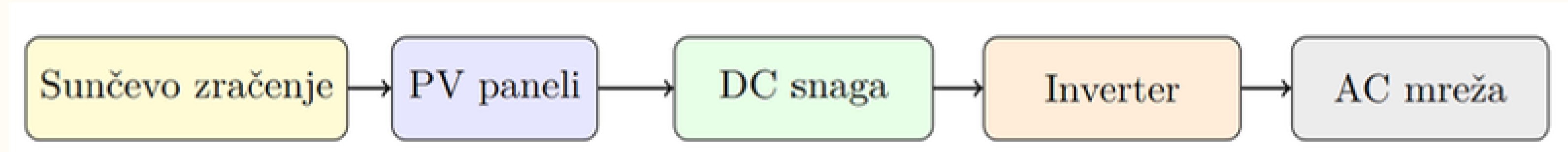




Solarna Elektrana + *inverter*

Ideja projekta

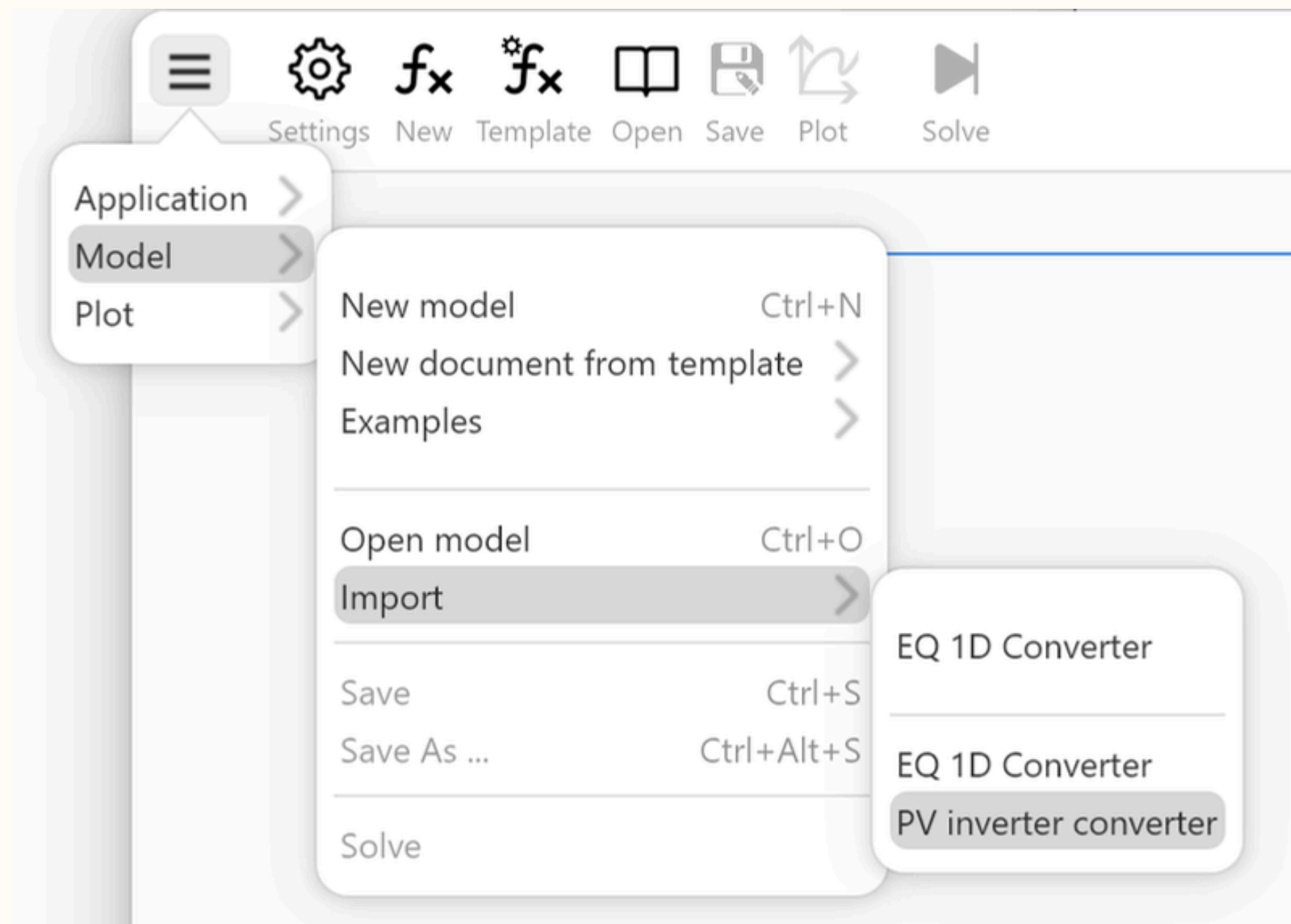
XML konfiguracija
GUI plugin
Preview proračun
Generisanje dTwin modela
Grafovi nakon Solve komande



Princip *rada* solarne elektrane

Integracija *plugin-a* u dTwin

Model → Import → PV inverter converter



Plugin se pokreće direktno
iz dTwin okruženja
Korisnik kroz GUI bira XML
konfiguraciju, testni case,
scenario i izlazni model

GUI *i Tok rada*

Config XML:

Case XML:

Output:

Test cases:

Scenarios:

Manual:

Status:

...

...

...

case9

case30

case118

case300

Sunny

Cloudy

Hot

Low PF

Bus: G: T: PF:

Recommend bus

Select config XML and case XML.

1. PROJECT AND BUS

Project and case selection

Test system: not selected

Buses: -

Branches: -

Base MVA: -

Connection bus

Selected bus: 5

Bus type: PQ/load

Validation: waiting

Bus summary

Bus Type Vm P Q

1 Slack 1.040 - -

5 PQ 1.000 125.0 50.0

2. CONNECTION PREVIEW

Connection preview

Select case and press Preview.

Before: original test case

After: PV inverter connected to selected bus

3. PV AND INVERTER

PV and inverter parameters

Solar plant

Rated DC power: from XML

Irradiance G: scenario/manual

Temperature Tcell: scenario/manual

Temperature coeff.: from XML

Inverter

Rated AC power: from XML

Power factor PF: scenario/manual

Reactive mode: fixed PF

Scenarios: Sunny, Cloudy, Hot, Low PF

4. RESULTS

Results at selected bus

Pdc available - MW

Pac injected - MW

Q ac injected - MVar

S apparent - MVA

PF actual -

Estimated V before - p.u.

Estimated V after - p.u.

Estimated delta V - p.u.

Per-unit values will be shown after Preview.

5. MATRIX INFORMATION

Matrix information

Dense Ybus: waiting

Sparse Ybus: waiting

Nonzeros: waiting

Backend: natID dense + sparse

6. PROGRESS AND LOG

[ready] select XML files or press a quick case button

Preview

Convert

Convert + p.u.

Ulazi:

Config XML, Case XML, Output

Podešavanja:

case, scenario, bus, G, T, PF

Akcije:

Preview, Convert, Convert + p.u.

Preview računa radnu
tačku

Convert generiše dTwin
model

Convert + p.u. dodaje
normalizovani prikaz

Preview *rezultati*

case30, bus 8, Cloudy scenario

Config XML:

C:\Users\Surface\OneDrive\Documents\SREES Solarna elektrana + inverter\src\config\solar_inverter.xml

—

Case XML:

C:\Users\Surface\OneDrive\Documents\SREES Solarna elektrana + inverter\src\cases\case30.xml

—

Output:

C:\Users\Surface\OneDrive\Documents\SREES Solarna elektrana + inverter\docs\generated_examples\case30_from_plugin.dmc

—

Test cases:

case9

case30

case118

case300

Scenarios:

Sunny

Cloudy

Hot

Low PF

Manual:

Bus: 8G: 400T: 20PF: 0.95Recommend bus

Status:

Preview completed.

1. PROJECT AND BUS

Project and case selection

Case: case30

Buses: 30

Branches: 41

Base power: 100 MVA

Connection bus

Selected bus: 8

Bus type: PQ/load

Validation: OK

Bus summary

Bus Type Vm P Q

1 Slack 1.040 - -

8 PQ 1.000 high load SELECTED

Full case loaded: 30 buses

2. CONNECTION PREVIEW

Connection preview (case30)

Original network

buses: 30

branches: 41

base: 100.00 MVA

Added element

type: grid-tied PV inverter

bus: 8 (PQ/load)

Pac: 3.26 MW

Qac: 1.07 MVar

PF: 0.95

Before: case30

After: case30 + PV inverter

Status: connection point validated

3. PV AND INVERTER

PV / inverter parameters

DC rated: 8.00 MW

AC rated: 7.20 MW

G: 400.00 W/m2

Tcell: 20.00 C

PF: 0.95

Typical daily profile

06:00 G=100 Pac=0.33 MW

09:00 G=500 Pac=1.63 MW

12:00 G=1000 Pac=3.26 MW

15:00 G=700 Pac=2.28 MW

18:00 G=150 Pac=0.49 MW

dTwin plots: MW/MVar + optional p.u.

4. RESULTS

Results at selected bus 8

Pdc available 3.2640 MW

Pac injected 3.2640 MW

Q ac injected 1.0728 MVar

S apparent 3.4358 MVA

PF actual 0.9500

Estimated V before 0.9700 p.u.

Estimated V after 0.9900 p.u.

Estimated delta V +0.0200 p.u.

Per-unit

Pac 0.4533 Q ac 0.1490 Pdc 0.4080

Warning: none

Status: preview

5. MATRIX INFORMATION

Backend:

natID dense:DbiMatrix + sparse:IDbiMatrix

Dense Ybus:

30 x 30

6. PROGRESS AND LOG

[ready] select XML files or press a quick case button

[info] Scenario selected: G=950 W/m2, T=32 C, PF=0.98.

[info] Case selected: case30.

[info] Scenario selected: G=400 W/m2, T=20 C, PF=0.95.

[info] Preview calculated successfully.

Preview

Convert

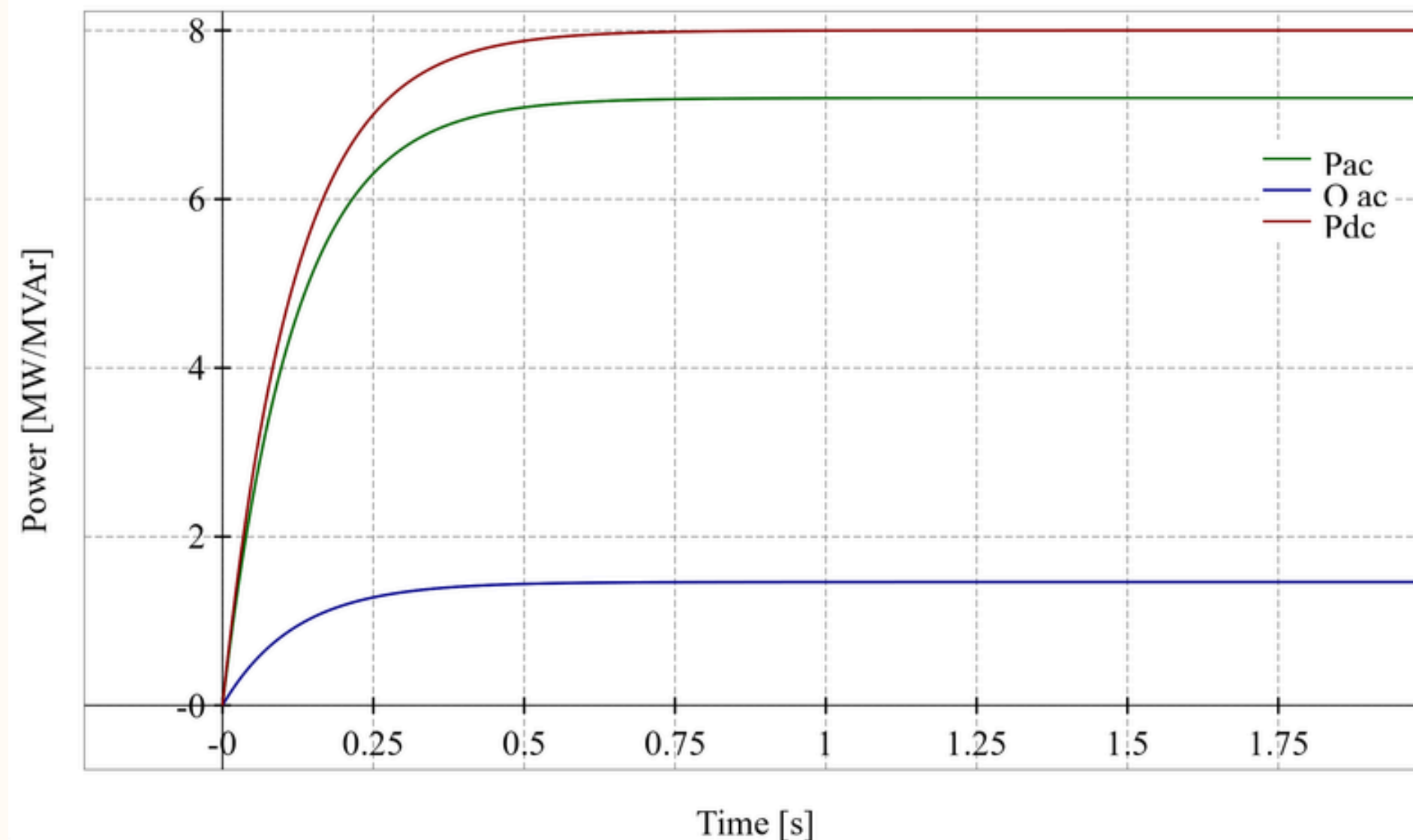
Convert + p.u.

G = 400 W/m²
T = 20 °C
PF = 0.95
Pdc = 3.264 MW
Pac = 3.264 MW
Qac = 1.073 MVar

Odabrani bus je
PQ/load tipa i
validacija je
uspješna.

Generisani dTwin model i simulacija

Dinamički DAE model prvog reda



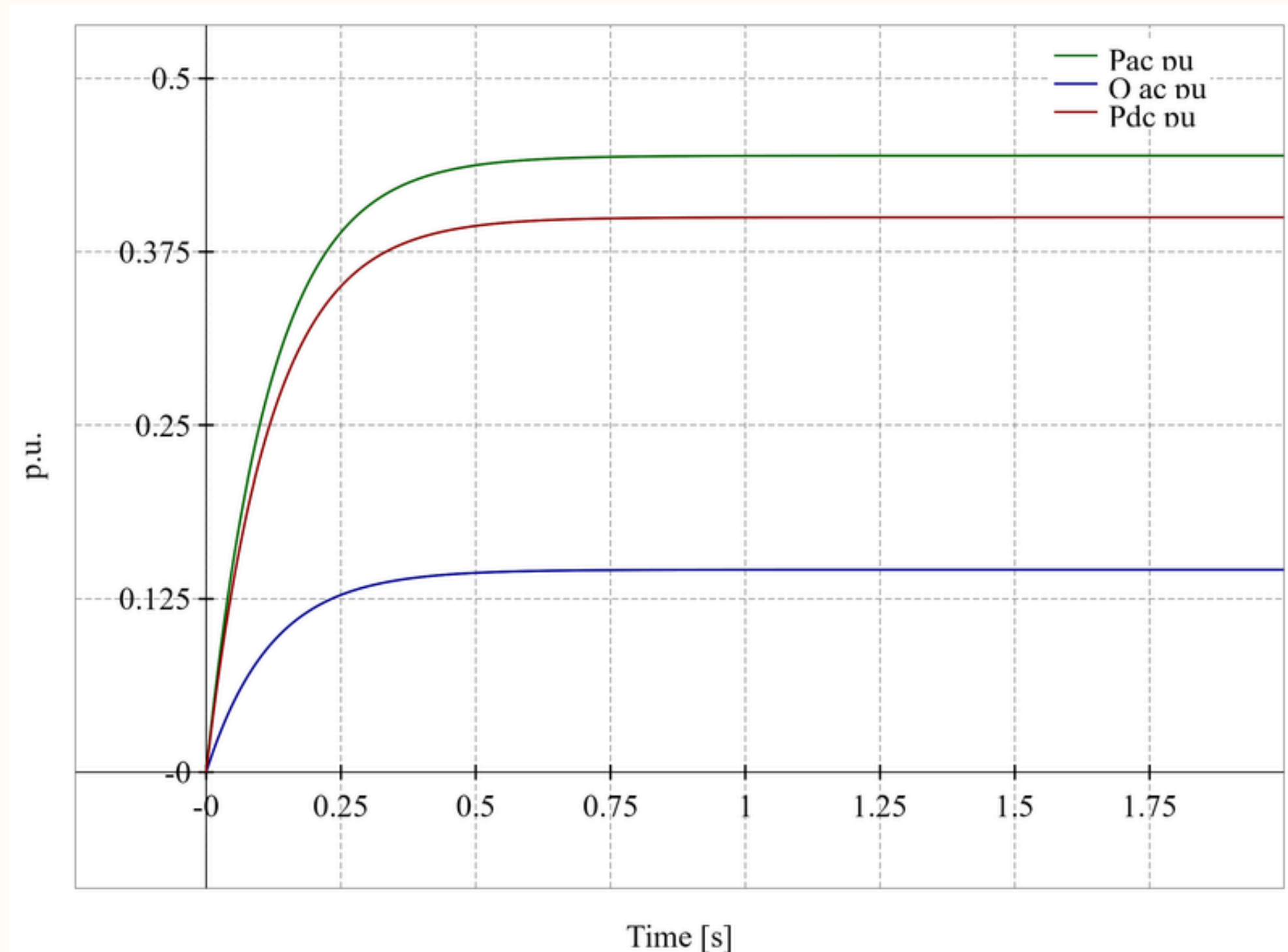
Pac prati referentnu aktivnu snagu
Qac prati referentnu reaktivnu snagu
Pdc prati dostupnu DC snagu

```
Model [type=DAE domain=real method=RK2 name="PV inverter model"]:  
Vars [out=true]:  
    p_ac = 0  
    q_ac = 0  
    p_dc = 0  
    irradiance = 400  
    cell_temp = 20  
Params:  
    p_ac_ref = 3.264  
    q_ac_ref = 1.07282  
    p_dc_ref = 3.264  
    inverter_tau = 0.12  
    connection_bus = 8  
    dc_rated_mw = 8  
    ac_rated_mw = 7.2  
    power_factor = 0.95  
    dense_rows = 30  
    dense_cols = 30  
    sparse_rows = 30  
    sparse_cols = 30  
    sparse_nonzeros = 89  
ODEs:  
    p_ac' = (p_ac_ref - p_ac) / inverter_tau  
    q_ac' = (q_ac_ref - q_ac) / inverter_tau  
    p_dc' = (p_dc_ref - p_dc) / inverter_tau  
    irradiance' = 0  
    cell_temp' = 0  
end
```

Realizovano:

XML konfiguracija
dTwin import plugin
izbor case-a i bus-a
model PV invertera
dTwin DAE model
grafovi u MW/MVAr i p.u.
natID dense/sparse backend

Projekat prikazuje kompletan tok od konfiguracije PV invertera do generisanog dTwin modela i simulacionih rezultata.



Rezultati *i zaključak*